AP PRECALCULUS

Unit 1 • Lesson 1.10 practice

Lesson 1.10 • Rational Functions and Holes

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Where is the hole in the graph of $r(x) = \dfrac{x^2 - 25}{x - 5}$?

(A) $(5, 0)$
(B) $(5, 10)$
(C) $(-5, 0)$
(D) $(5, 5)$

Question 2

NO CALCULATOR

Where is the hole in the graph of $r(x) = \dfrac{x^2 + x - 2}{x^2 - 1}$?

(A) $\left(1, \frac{3}{2}\right)$
(B) $(-1, 0)$
(C) $(1, 0)$
(D) $(-2, 0)$

Question 3

NO CALCULATOR

What happens on the graph of $r(x) = \dfrac{x - 2}{(x - 2)^2}$ at $x = 2$?

(A) A hole
(B) A zero
(C) A vertical asymptote
(D) Nothing; r is defined at $x = 2$.

Question 4

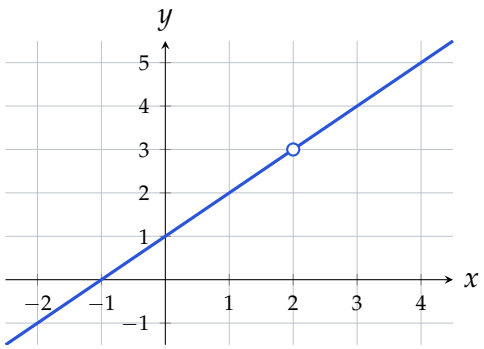
NO CALCULATOR

$\lim\limits_{x \to 3} \dfrac{x^2 - 3x}{x - 3}$ is

(A) 0
(B) 1
(C) 9
(D) 3

Question 5 (graph) NO CALCULATOR

The graph of a rational function r is shown: a line with a hole at $(2, 3)$. Which could be $r(x)$?



(A) $\dfrac{x^2-4}{x-2}$

(B) $\dfrac{x+1}{x-2}$

(C) $\dfrac{x^2-x-2}{x-2}$

(D) $x+1$

Question 6 (Free response) NO CALCULATOR

Let $r(x) = \dfrac{x^3-4x}{x^2+2x}$.

(a) Factor $r(x)$ completely and simplify.

(b) Find the coordinates of every hole in the graph of r.

(c) Does the graph of r have a vertical asymptote? Find the zeros of r and describe the graph.